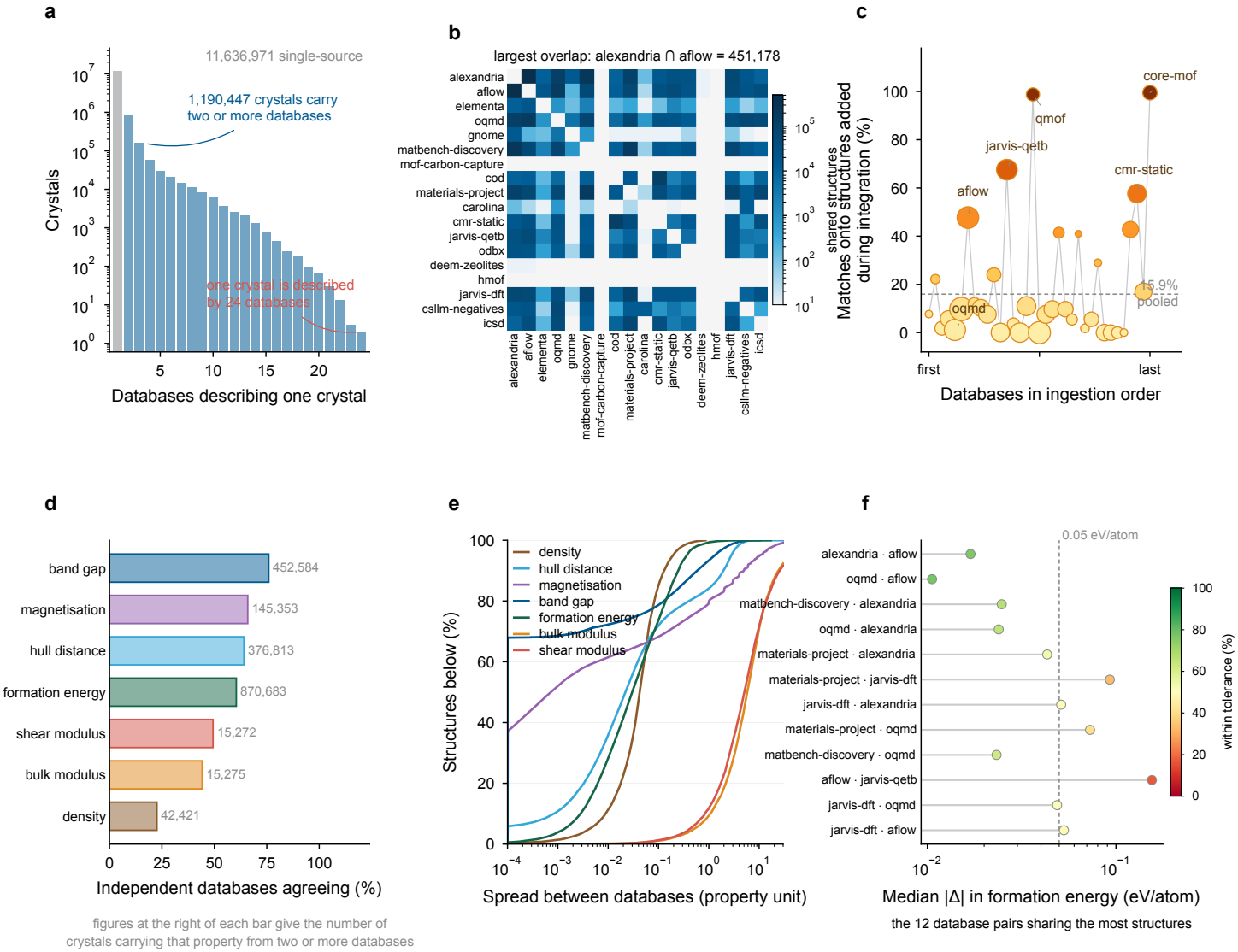

a
11,636,971 single-source
1,190,447 crystals carry two or more databases
one crystal is described by 24 databases
Crystals
Databases describing one crystal
b
largest overlap: alexandria ∩ aflow = 451,178
alexandria
aflow
elementa
oqmd
gnome
matbench-discovery
mof-carbon-capture
cod
materials-project
carolina
cmr-static
jarvis-qetb
odbx
deem-zeolites
hmof
jarvis-dft
csllm-negatives
icsd
shared structures
c
Matches onto structures added during integration (%)
core-mof
qmof
jarvis-qetb
cmr-static
aflow
oqmd
5.9% pooled
first
last
Databases in ingestion order
d
band gap
452,584
magnetisation
145,353
hull distance
376,813
formation energy
870,683
shear modulus
15,272
bulk modulus
15,275
density
42,421
Independent databases agreeing (%)
figures at the right of each bar give the number of crystals carrying that property from two or more databases
e
density
hull distance
magnetisation
band gap
formation energy
bulk modulus
shear modulus
Structures below (%)
Spread between databases (property unit)
f
0.05 eV/atom
alexandria · aflow
oqmd · aflow
matbench-discovery · alexandria
oqmd · alexandria
materials-project · alexandria
materials-project · jarvis-dft
jarvis-dft · alexandria
materials-project · oqmd
matbench-discovery · oqmd
aflow · jarvis-qetb
jarvis-dft · oqmd
jarvis-dft · aflow
within tolerance (%)
Median |Δ| in formation energy (eV/atom)
the 12 database pairs sharing the most structures